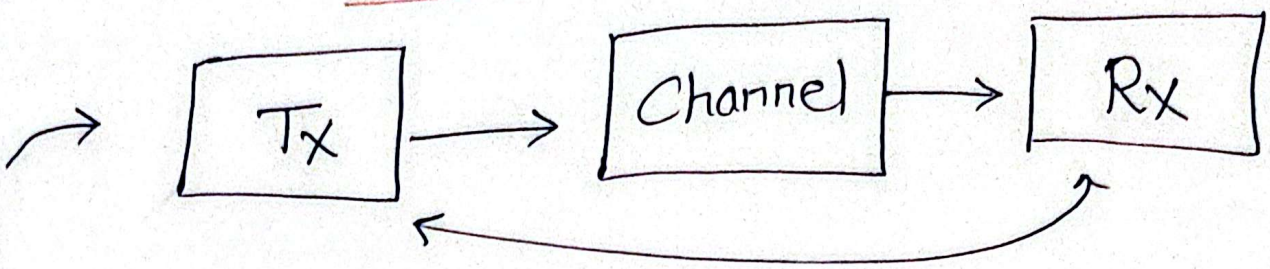
* Digital Communication *

SALEH RAMADAN

" Computer "

" Random Processes "

" Section 2 "



We need to Analze Them in "Digital Domain"

→ Before starting Analysis:

- Data ⟹ Random
- Noise ⟹ Random
- Interferance ⟹ Random
- Channel ⟹ Random

→ So we need to Know how to deal With This Randomness Through Studying

Random Variables

Random Processes.

CamScanner

* Probability:

① Identities:

- **Random Experiment:** Experiment whose output is not deterministic

- **Sample space:** "S" space will all Possible outcomes

- $P(S) = 1$

- **Event:** "W_i" Subset of outcomes of certain Feature

$$P(W_i) \geq 0 \quad , \quad P(W_i) < 1$$

- **Mutually Exclusive:** for Two Events A, B

$$P(A \cap B) = 0$$

A and B couldn't be happen in The Same Time

$$P(A \cup B) = P(A) + P(B)$$

CamScanner

(1)

- $P(\emptyset) = 0$, $P(\bar{A}) = 1 - P(A)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

---

(2) Conditional Probability: *Important*

- $P(A|B) = \frac{P(A \cap B)}{P(B)} \Longrightarrow$ كانها بتحدد مقدار الشبه بين A, B
- Baye's Rule: $P(B|A) = \frac{P(A|B) * P(B)}{P(A)}$

---

(3) Total Probability:

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots$$

---

*Note: - $P(A \cap B) = P(A)P(B) \Longleftrightarrow A, B$ are Independa

- $P(A|B) \Longrightarrow$ It's a Ratio Not Absolute Probability.

---

(4) Random Variables: Map outcomes of Experiment to Real Numbers

(4.1) Single RV:

* Types: Continous , Discrete
* Defintion: PMF "Probability Mass Function" , PDF "Probability Density Function"

* Functions: Probability ? PMF

① PMF: $P(X = x) = P(x)$ ← فالكورس هنا الـ PDF والـ PMF مسميينهم $f(x)$ بكتب الصح إنها $P(x)$

- $P(x) \geq 0$, $P(x) < 1$
- $\sum P(x) = 1$

② PDF: Probability مش density بتعبر عنه.

→ For Continous RV: $P(X = a) = 0$

- $P(a \leq x \leq b) = \int_a^b f(x)\,dx$
- $f(x) \geq 0$, "$f(x)$ can be > 1"
- $\int f(x)\,dx = 1$

③ CDF: Cummulative Distribution Function

- For Discrete RV: $F(x) = P(X \leq a) = \sum_{x \leq a} P(x)$
- For Continous RV: $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt$
- $P(a < x < b) = F(b) - F(a)$
- $f(x) = \frac{dF(x)}{dx}$

CamScanner

④

- $0 \leq F(X) \leq 1$
- $\lim_{X \to -\infty} F(X) = 0$
- $\lim_{X \to \infty} F(X) = 1$

} Monotone Non-decreasing Function

* Expectation & Variance:

$Var(X) = E((X-\mu_X)^2)$

① Mean: $\mu_X = E(X)$, ② Variance: $Var(X) = \sigma_X^2 = E(X^2) - \mu_X^2$

$E(g(X)) = \int g(X) F(X)\, dX \Longrightarrow$ Continous RV

$E(g(X)) = \sum g(X) P(X) \Longrightarrow$ Discrete RV

Then:
- $E(X) = \int X f(X)\, dX$, $E(X) = \sum X P(X)$
- $E(X^2) = \int X^2 f(X)\, dX$, $E(X^2) = \sum X^2 P(X)$

* Notes:

Mean $\Longrightarrow$ DC

Then: $\sigma_X^2 = E(X^2) - \mu_X^2$

- σ_X^2 ↙ A.C Power
- $E(X^2)$ ⇓ Total Power
- μ_X^2 ⇓ DC Power

- $E(aX+b) = aE(X) + b$
- $Var(aX+b) = a^2 Var(X)$

CamScanner

*Popular Distributions

① Bernoulli : Discrete RV

$X \sim Bern(P)$

$P(X=1) = P$, $P(X=0) = 1-P$

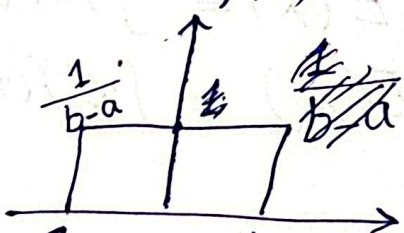
② Binomial: Discrete RV $X = 0, 1, 2, 3, \dots n$

$P_X(x) = {}^{n}C_{x} * P^{x} (1-P)^{n-x}$, $x = 0, 1, 2, \dots n$

n: Number of Trials

③ Uniform: Continous RV

$X \sim U(a,b)$

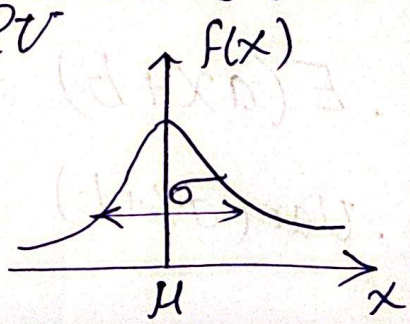


- $f(x) = \frac{1}{b-a}$; $a \leq x \leq b$
- $\mu_X = \frac{a+b}{2}$
- $\sigma_X^2 = \frac{(b-a)^2}{12}$

④ Gaussian (Normal): Continous RV

$X \sim N(\mu, \sigma^2)$

$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)$ $-\infty < x < \infty$



CamScanner

AWGN $\sim N(0, \sigma^2)$

4.2 Two RVs: here we define Joint Functions

* Functions:

① Joint PMF: $\cdot P(X=x, Y=y) = P(x,y)$

$\therefore \sum_x \sum_y P(x,y) = 1$

② Joint PDF: $\cdot P((x,y) \in A) = \iint_A f(x,y)\,dx\,dy$

$\cdot \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1.$

③ Joint CDF: $F(x,y) = P(X \leq x, Y \leq y)$

* Marginal Distributions:

$\cdot P_X(x) = \sum_y P(x,y)$, $P_Y(y) = \sum_x P(x,y)$

$\cdot f_X(x) = \int_y F(x,y)\,dy$, $f_Y(y) = \int_x f(x,y)\,dx.$

CamScanner

* Covariance & Correlation:

$$Cov(X,Y) = E((X-\mu_X)(Y-\mu_Y))$$

$$\sigma_{XY} = E(XY) - \mu_X \mu_Y$$

- $\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$ → 1, 0 ⟹ uncorrelated, −1

* Independance & Correlation & orthogonality:

→ Conditional PMF & PDF:

$$f_{X|Y}(x,y) = \frac{f(x,y)}{f(y)} \quad , \quad P_{X|Y} = \frac{P(x,y)}{P_Y(y)}$$

** $f_{X,Y}(x,y) = f_X(x)\, f_Y(y) \iff$ X, Y are Independant RVs

$$f_{X|Y}(x,y) = f_X(x) \quad , \quad E(XY) = E(X)\,E(Y)$$

→ $Cov(X,Y) = 0 \iff$ X, Y are uncorrelated

Independant ⊂ uncorrelated

uncorrelated ⊂ Independant

If X, Y are Independant $\Longrightarrow$ uncorrelated

But If X, Y are uncorrelated $\Longrightarrow$ Independant

Not all Times

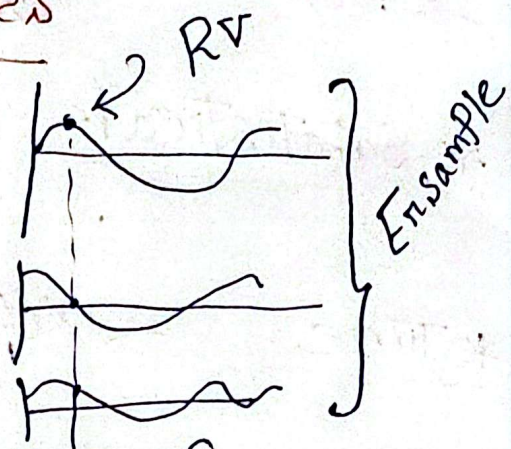
only For Gaussian Distribution.

$\longrightarrow$ $E(XY) = 0$ $\Longleftrightarrow$ X, Y are orthogonal

SALEH RAMADI

"Random Processes"

"Summary"



* Random Process:

→ Definition: Mapping outcomes of Random Experiment to Functions of Time

each Function is Called ⟹ "Realization"

All Functions are called ⟹ "Ensample"

- Sampling of RP @ Time Instant ⟹ "Random Variable"

→ Parameters:

① Expectation:

* Satatistical: Across all Realization

- $m_x(t) = E(x(t))$

* Time: Across Time for one Realization

$$\langle x(t) \rangle = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} x(t)\, dt$$

CamScanner

② Auto correlation:

→ Definition: $R_x(t_1, t_2) = E(x(t_1) \cdot x(t_2))$

* Satatistical: $R_x(t_1, t_2) = E(x(t_1) \cdot x(t_2))$

* Time: $\langle x(t)\, x(t+\tau) \rangle = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} x(t)\, x(t+\tau)\, dt$

---

③ Power spectral Density: (PSD)

$S_x(f) = F.T(R_x(\tau))$

→ Types:

RP

- Non-Stationary → Statatistics are varying
- Stationary → Statatistics are constant

* Stationary RPs:

① Strict Sense Stationary (SSS):

① ALL RVs are Identically distributed
② ALL Joint PDFs depend only on Time shift

* Wide Sense Stationary (WSS):

(1) Mean ($m_x(t)$) is constant

(2) Autocorrelation ($R_x(t_1, t_2)$) is Function in Time shift ($R_x(\tau)$)

→ Properities of $R_x(\tau)$:

- $R_x(\tau) = E(x(t)\, x(t+\tau))$
- $R_x(\tau) = + R_x(-\tau)$ "Even Function"
- $R_x(0) = E(x^2(t)) =$ Total Average Power

$$R_x(0) = \int_{-\infty}^{\infty} S_x(f)\, df$$

- $R_x(\infty) = [E(x(t))]^2 = \mu^2 =$ D.C. Power
- $R_x(0) \geq R_x(\tau)$, $\tau \neq 0$

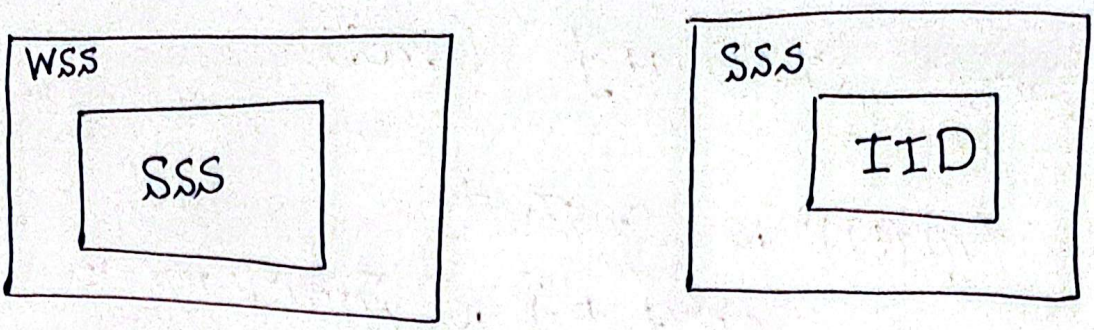
* Note: IID Process

(1) ALL RVs have Identical PDFs

(2) ALL RVs are Independant.

CamScanner

(4)

* 

* Ergodic Process:

① Time Mean = Stat. Mean

$$m_x(t) = \langle x(t) \rangle$$

② Time ACF = Stat. ACF

$$\langle x(t)x(t+\tau) \rangle = E(x(t)\,x(t+\tau))$$

* LTI Filtering of RP:

$x(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t)$

$x(t)$: WSS

$m_x(t)$

$R_x(\tau)$

$S_x(f)$

$y(t)$: WSS

* $m_y(t) = m_x(t)\,H(0)$

* $R_y(\tau) = h(-\tau) \circledast h(\tau) \circledast R_x(\tau)$

* $S_y(f) = |H(f)|^2 S_x(f)$

CamScanner